BASANT BHATTARAI

SENIOR DATA ENGINEER / AI ENGINEER / DATA & AI ARCHITECT

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PROFILE

Senior data engineer and data and AI architect with 9+ years on production systems in industry. At UXCam, owns the architecture of a 15 TB+ data platform processing 100M+ events a day and the production AI agents built into it: 40% lower infrastructure cost, 50% faster processing, 60% faster analytics delivery, 99.9% uptime. Sets the target architecture, chooses platforms on measured evidence, designs the data model with its governance and SOC 2 controls, and leads the build on Kafka, Spark, Iceberg, Trino, dbt and Airflow on Kubernetes. Open to Senior or Lead Data Engineer, AI Engineer and Data & AI Architect roles.

EXPERIENCE

Senior Data Engineer | UXCam

Feb 2024 - present

- Own the architecture, reliability, data quality and governance of a 15 TB+ platform processing 100M+ events daily: freshness and latency objectives per dataset, monitoring and alerting, 99.9% uptime, incident-ready runbooks, reviewable data contracts, a named owner for every published table, and the SOC 2 security criteria met for the platform and its infrastructure through every audit.
- Design and operate the production agent workflows — LangGraph and Google ADK orchestration, MCP tool servers, retrieval and evaluation — built into the existing platform, so model output lands as typed, provenanced rows rather than a transcript: analytics delivery 60% faster with 75% less analyst effort, review and fallback paths kept.
- Run the architecture and schema review every new store, pipeline and agent workflow passes before production, and operate the SQL and NoSQL databases underneath: replication, recovery, access, tuning, upgrades.
- Mentor 5 engineers through pairing, code review and written design feedback; team delivery time down 30%.

Data Engineer | UXCam

Feb 2020 - Feb 2024

- Rebuilt the core ETL layer in Spark and PySpark with Airflow orchestration, cutting processing time for 5 TB+/day by 50% and unblocking real-time product dashboards.
- Made backfills and replays safe to re-run: idempotent loads, explicit watermarks for late data, and schema evolution designed in.
- Tuned Trino, ClickHouse, Citus and TimescaleDB serving 100M+ queries/day for a 50% p95 latency reduction through partitioning, clustering and model redesign.
- Led the Kubernetes migration of the data and database infrastructure: 40% lower infrastructure cost and autoscaling that held under peak load.

Project Leader and Research Developer | SVCET, JNTU Anantapur

Nov 2018 - Mar 2019

- Final-year research project: an intelligent dialogue system on deep learning and NLP, a sequence-to-sequence model of two RNNs in TensorFlow and Python, trained on movie-dialogue and Twitter corpora, with a Flask web interface; led 4 developers and delivered 2 weeks ahead of schedule.

Software Developer Intern (part-time) | SV Technology, India

2017 - 2019

- Built Python backend services in Flask and Django serving 10K+ users, at 85% test coverage, alongside the degree.
- Designed and maintained PostgreSQL schemas, stored procedures and reporting paths, and set up CI with Jenkins alongside early pipeline work, the start of the move into data engineering.

SELECTED PROJECTS

- **Database scale-out and migration:** Postgres application databases and a pgvector store moved to Citus as they outgrew one server; an RDS database moved to a self-managed Citus cluster; a ClickHouse cluster designed to its workload.
- **Reliability and cost:** PostgreSQL and TimescaleDB clusters with replication, pgBackRest and PgPool-II; a database recovered from its raw files; an RDS and Aurora fleet tuned and failover-tested; AWS database spend analysed for savings.

- **Data platforms and AI:** an analytics lakehouse on S3, Trino, Airflow and Glue, delivered running with access control, HA and DR; frameworks for LLM applications and agentic workflows, architecture to tested Python; a failing classifier rebuilt on Databricks.

SELECTED OPEN SOURCE

- **data-platform-architecture** and **ai-platform-architecture** (reference architectures): a SaaS company leaving Postgres, measured: 18 decision records, a Kimball model, SOC 2 readiness, a cost model and a Databricks proof of concept; then its usage assistant on governed data.
- **spark-sql-kinesis** (maintainer): Kinesis source and sink for Spark 4.2 on AWS SDK v2, on Maven Central; an idle shard costs one GetRecords call, not a multi-second floor per batch.
- **governed-agent** and **evalproof** (LangGraph, MCP, LLM evaluation): answers matched a SQL-computed key on 48/54 questions, and 0 of 5 planted prompt injections led to a proposed action; evalproof fails CI only on LLM-judge drops both large and significant.
- Also **distributed-db-lab** (Citus sharding and online rebalance, against TimescaleDB and ClickHouse), data-stack-lab, streaming, real-time ML, retrieval, data contracts and GitOps delivery. github.com/technobasant

SKILLS

Data and AI architecture

Target architectures and platform selection, conceptual to physical modelling, data warehouse and lakehouse modelling (Kimball, Type 2 history), ClickHouse and Citus layouts, contracts in CI, decision records and architecture documentation, SOC 2 and GDPR erasure by design, tenant isolation outside the model, semantic layers, release gates, cost models.

Data platforms

Python, SQL and Scala; Kafka and Kinesis ingestion (maintainer of a Spark 4 Kinesis connector); Spark/PySpark processing, including the Spark 4 migration; Airflow orchestration; Iceberg, Hive Metastore, dbt and lakehouse modelling; Databricks with Unity Catalog; CI/CD with GitHub Actions, Kubernetes, Docker and Terraform; data quality, contracts, lineage, cost and recovery; observability with Prometheus and Grafana.

AI engineering

LangGraph, LangChain, Google ADK, MCP servers, Pydantic, Milvus and MLflow; prompt and structured-output design, bounded tools, retrieval, evaluation before rollout on Qwen3, Llama 3.3, gpt-oss and Databricks Genie, provenance, human review and honest fallback paths; deep-learning NLP in TensorFlow since 2018.

Database lifecycle

PostgreSQL, ClickHouse, Trino, Citus, TimescaleDB, Redis and MongoDB; partitioning, replication, restore and high availability.

Cloud

AWS (production): S3, EKS (IRSA), Lambda, Glue, Athena, Kinesis, EMR, RDS | **GCP** (working knowledge): BigQuery, Dataflow, Dataproc, Datastream, Pub/Sub | **Azure** (working knowledge): Synapse, Data Factory, Databricks, ADLS, Event Hubs

EDUCATION AND LANGUAGES

Bachelor of Technology in Computer Science and Engineering

JNTUA College of Engineering, Anantapur, India | 2015 - 2019

English: C1 | Nepali: native

Selected work, writing and case studies: basantbhattarai.com.np/work/